

**DEEP LEARNING-BASED DETECTION OF MANGROVE GAIN
AND LOSS IN ARINGAY, LA UNION**

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APPROVAL SHEET

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DEDICATION

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ABSTRACT

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Mangrove ecosystems play an important role in protecting coastal areas, supporting biodiversity, and maintaining environmental balance. However, changes in mangrove areas caused by natural and human activities may affect their ecological functions and sustainability.

This study aims to develop a deep learning-based image processing system for detecting and quantifying mangrove gain and loss in Aringay, La Union. Specifically, it seeks to collect and prepare a localized mangrove dataset using satellite imagery and utilize the U-Net deep learning architecture to train a model capable of identifying changes in mangrove cover over time. The developed model will then be evaluated based on its performance using F1-score, Intersection over Union (IoU), mean Average Precision to determine its effectiveness in detecting mangrove gain and loss.

The proposed study is intended to provide an automated approach for identifying changes in mangrove areas and generate information that may assist in environmental monitoring and conservation planning. The results of the study may serve as a basis for future research and support the use of deep learning and remote sensing technologies in monitoring mangrove ecosystems in Aringay, La Union.

Keywords: *change detection, deep learning, mangrove gain and loss, mangrove monitoring, remote sensing*

Chapter 1

INTRODUCTION

Situation Analysis

Deep learning has evolved into an extensive tool that can be used for pattern recognition in large and complex data, especially imagery data, in a variety of fields. The use of Convolutional Neural Network (CNN) for automatic feature extraction from imagery has become a common practice in these fields. For example, the use of deep learning for agricultural purposes, such as crop classification, diseases detection, and yield monitoring via remotely sensed imagery has been proved through various research in the past (Zhang et al., 2025). In a bibliometric analysis carried out on more than 650 papers from 2000 to 2024, it was found that CNNs were the leading technology used in real-time monitoring of crops, while problems like the lack of annotated data and the limitations of models' generalization were found to be prevailing (Zhang et al., 2025). Such difficulties apply not only to agricultural purposes but also to other environmental monitoring tasks.

Deep learning and remote sensing together expand applications beyond image classification to include land cover classification and ecosystem monitoring. Satellite missions such as Sentinel-2 have proven useful for environmental monitoring applications because they offer freely available, recurrent imaging (ESA, 2024). Creating a unified remotely sensed product across a wide geographic range has been accomplished via global initiatives like the Global Mangrove Watch (Bunting et al., 2022). Semantic segmentation models, specifically the U-Net architecture, have been adopted for pixel-level

environmental classification due to the ability of its encoder-decoder architecture with skip connections to preserve fine detail (Ronneberger et al., 2015).

Among the various environmental monitoring tasks, monitoring of mangrove forests has received considerable interest. Mangroves serve the purpose of coastal defense, provide biodiversity habitats, and store carbon. The extent of the global mangrove cover has decreased significantly, as indicated by Worthington et al. (2020) and Food and Agriculture Organization of United Nations (2023). However, mangrove loss has been particularly acute in Southeast Asia because of extensive regional deforestation activities and the percentage share of mangroves globally situated in the same region (Lomeo & Singh, 2022).

In the Philippines, past estimations put the extent of national mangroves at between 400,000 ha to 500,000 ha, whereas modern estimates have shown considerable reductions (Pillodar et al., 2023; Sasi et al., 2025). Mangrove depletion in the Philippines has previously been attributed to aquaculture expansion, land reclamations and other forms of change in land use. However, the actual quantities and underlying causes of changes differ depending on regional contexts (Primavera, 2000). This variability suggests the necessity for detailed spatial information on mangrove distributions and changes at the barangay scale. General national and global databases might not capture localized variations accurately enough. Coastal barangays within Aringay, La Union, such as Barangay Dulao, constitute mangrove resources of the Ilocos Region. There is currently no localized and automated mangrove monitoring scheme available for this region.

There are examples of utilizing deep learning semantic segmentation models with Sentinel-2 imagery for mapping mangroves, Maung et al. (2024) utilized a U-Net model

with multisource remote sensing data for the detection of mangrove areas, whereas Jamaluddin & Chen (2024) evaluated various deep learning semantic segmentation frameworks with Sentinel-2 imagery for mangrove detection activities in Florida. Prior studies conducted in Bohol, Philippines have demonstrated that the baseline U-Net model excels in semantic segmentation for local mangrove mapping using Sentinel-2 images (Auxtero & Villamor, 2026). While more recent architectures, including transformer-based models, have demonstrated superior performance over U-Net in some mangrove-monitoring comparisons (Panuntun et al., 2024), these approaches typically require larger and more diverse training datasets than are currently available for Barangay Dulao. U-Net remains a well-established and appropriately scaled architecture for localized, pixel-level mangrove segmentation, particularly given its demonstrated effectiveness in comparable Philippine coastal settings (Auxtero & Villamor, 2026) and its lower data and computational requirements relative to more complex alternatives.

Prior studies have already demonstrated the effectiveness of using Sentinel-2 multi-temporal images to evaluate mangrove degradation and changes in extent (Farzanmanesh et al., 2024; Jamaluddin et al., 2021). Vegetation indices such as Normalized Difference Vegetation Index (NDVI) have already been evaluated for enhancing the distinction between mangrove vegetation and nearby areas (Maung et al., 2024; Xue & Qian, 2022). The consistency of timing in satellite observations has been shown to significantly impact the accuracy of classification performance across consecutive observation periods (Valderrama-Landeros et al., 2021). Global and national datasets can serve as valuable resources, offering pertinent information related to mangrove ecosystems. Nonetheless, the low resolutions used, and general classification methods applied may restrict the capacity

to identify localized transitions, fragmentations, or narrow coastlines within Barangay Dulao.

While previous studies have shown the success of deep learning in agricultural and environmental remote sensing applications, and recent research focused on mangroves has highlighted the potential of Sentinel-2 imagery and U-Net-based semantic segmentation for mapping these ecosystems, there is still a need for a localized dataset and a combined approach for detecting gains and losses in the mangrove areas of Barangay Dulao, Aringay, La Union. Current global and regional datasets can offer valuable reference data; however, they might not fully capture localized mangrove conditions and the temporal variations within the research area.

This gap emphasizes the necessity for the current study, which aims to gather a localized mangrove dataset and create a U-Net-based model for detecting mangrove gain and loss utilizing Sentinel-2 multispectral imagery. The evaluation of the model will utilize F1-score, Intersection over Union (IoU), and mean Accuracy Precision (mAP), assessing model performance relative to the outcomes of the initial trained iteration as a baseline instead of a predetermined acceptance threshold. The classifications produced will serve to pinpoint and measure the mapped increase and decrease of mangroves from 2019 to 2024.

Statement of Objectives

The main objective of this study is to develop an image processing system that automatically detects and quantifies mangrove cover change over time in Aringay, La Union using satellite imagery and deep learning-based segmentation.

Specifically, this study aims:

1. To collect a localized mangrove dataset.
2. To train the mangrove detection of gain and loss model using U-Net.
3. To evaluate the performance of the mangrove detection of gain and loss model in term of;
 - a. F1-score;
 - b. Intersection over Union (IoU); and
 - c. mean Average Precision (mAP).

Time and Place of the Study

The study will be conducted along the shoreline of Barangay Dulao, Aringay, La Union. This site was selected due to its confirmed coastal mangrove presence within the intertidal zone, as verified through Normalized Difference Vegetation Index (NDVI) analysis of Sentinel-2 imagery. The absence of a localized, automated mangrove monitoring system in the area makes it an ideal setting for deploying a deep learning-based change detection framework to support the Aringay MENRO in tracking mangrove cover changes over time.

Definition of Terms

For a better understanding of this research report, the following terms are conceptually and operationally defined:

Change Detection refers to the process of identifying differences in mangrove cover by comparing satellite imagery acquired at two or more points in time.

Convolutional Neural Network (CNN) refers to a deep learning architecture that automatically extracts spatial features from image data.

Deep Learning refers to a subset of machine learning that employs multi-layered neural networks to recognize complex patterns.

Image Processing System refers to a software system that applies computational methods to analyze and interpret satellite imagery.

Mangrove Cover refers to the total area occupied by mangrove vegetation within the study site expressed in hectares.

Mangrove Gain refers to an increase in mangrove cover between two time points, identified in this study when pixels classified as non-mangrove in an earlier year are reclassified as mangrove in a succeeding year.

Mangrove Loss refers to a decrease in mangrove cover between two time points, identified in this study when pixels classified as mangrove in an earlier year are reclassified as non-mangrove in a succeeding year.

Mangrove Segmentation refers to the pixel-level classification of satellite imagery into mangrove and non-mangrove classes.

Mean Average Precision (mAP) refers to a threshold-independent evaluation metric computed as the area under the precision–recall curve of the model's probability map. In this study, because the segmentation contains a single positive class (mangrove), mAP is equivalent to the Average Precision of the mangrove class.

Normalized Difference Vegetation Index (NDVI) refers to a vegetation index derived from the near-infrared (NIR) and red spectral bands that represents their normalized reflectance difference. In this study, NDVI is used as an additional input feature

of the U-Net model together with Sentinel-2 Bands 3, 4, 8, and 11 to support the identification of mangrove and non-mangrove areas.

Precision–Recall Curve refers to a plot of precision against recall obtained by sweeping the classification threshold across the model's probability map, where each threshold yields one confusion matrix and one precision–recall pair.

Satellite Imagery refers to digital multispectral images of the Earth's surface captured by orbiting satellites.

Segmentation Model refers to a deep learning model that assigns a class label to every pixel in an input image.

Sentinel-2 refers to a freely available European Space Agency satellite that provides multispectral imagery at 10-meter spatial resolution.

U-Net refers to an encoder-decoder convolutional neural network architecture with skip connections that preserve fine spatial detail, used in this study as the primary segmentation model for detecting mangrove boundaries in Sentinel-2 imagery.

Validation refers to the process of comparing model-generated outputs against reference datasets and ground records to evaluate accuracy.

Chapter 2

METHODOLOGY

Research Design

This study will be classified as a hybrid experimental-descriptive research design, integrating the development and evaluation of a deep-learning model with a descriptive account of resulting land-cover change. The experimental component will involve preparing Sentinel-2 multispectral imagery and reference masks, training a U-Net convolutional neural network for pixel-level binary segmentation of mangrove and non-mangrove areas, and evaluating the trained model using Intersection over Union (IoU), F1-score, and mean Average Precision (mAP).

Model refinement will occur only during the model-development stage: rather than requiring a fixed acceptance threshold, the performance of the first trained model will serve as the baseline for subsequent iterations, with each later iteration evaluated against this baseline and against preceding iterations as researcher-defined evaluation criteria. Should an iteration fail to improve upon the baseline, the researchers will examine possible sources of underperformance among the training data, preprocessing, augmentation, learning parameters, class-imbalance treatment, or model parameters. Reference-label adjustments will follow the established labeling procedure and will not be made solely for performance reasons.

After the final model configuration is chosen, it is fixed before application to the multi-temporal imagery; since the study does not create a truly independent test dataset, validation results are considered estimates for the reference-data conditions for the study.

The descriptive component will be completed after model finalization, where annual mangrove/non-mangrove maps (2019–2024) will be compared to one another over consecutive years to determine mapped mangrove gain, loss, and stability, and areas mapped for each category.

Detected transitions will be interpreted as probable land-cover changes rather than confirmed ecological causes, since factors such as erosion, tidal conditions, cloud contamination, or classification error may also produce similar mapped outcomes; where supporting local records or imagery exist, detected transitions will be examined alongside them for context.

Materials and Procedures

This study will use Sentinel-2 satellite images, reference data, and image processing tools. A U-Net model will be used to classify the images. The development will follow the Iterative and Incremental Development (IID) framework, organized into eight phases: data collection, image preprocessing, reference labeling, model training, change detection, prototype development, and system evaluation.

The IID framework will guide the development of the system through repeated cycles of building, testing, and improving (Larman & Basili, 2003; Ibrahim et al., 2020). The iterative part means that if the model's performance is not good enough, the researchers will go back and adjust the data or model settings. The incremental part means the system will be built in steps: first the data preparation, then the model, then the change detection, and finally putting it all together into a working prototype.

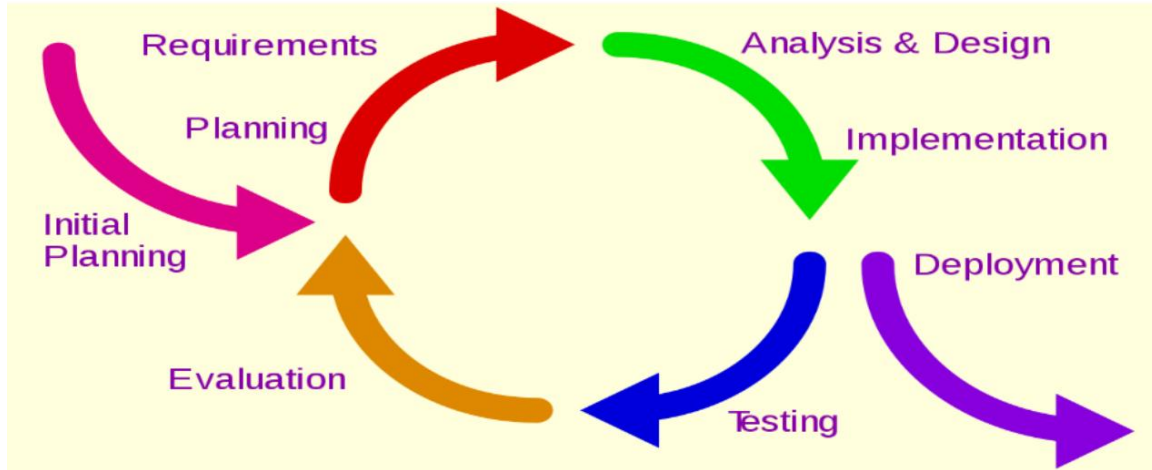


Fig. 2. Iterative and Incremental Development Model. Note. Adapted from Jevon (2009, as cited in Ibrahim et al., 2020).

The phases will occur in a defined order so that when model validation feedback indicates the requirement of refinement, the researchers would return to the correct prior phase. As soon as the final model configuration and classification threshold have been selected using the training and validation subsets, they will be locked, and the final model will be uniformly applied to the multi-temporal imagery.

Data Acquisition. In this phase, satellite imagery and reference data will be collected to train, evaluate, and analyze changes in the land cover over time. Satellite images of Barangay Dulao, Aringay, La Union will be retrieved for each year from 2019 through 2024. These images capture key color bands that help the computer vision model distinguish mangrove areas from non-mangrove vegetation and land.

A map boundary file will be used to define the exact study area. Image selection will prioritize full geographic coverage, minimal cloud cover, low shading, and high image quality. To ensure consistency across annual comparisons, imagery will be selected from

the same season each year to minimize variations caused by seasonal weather and plant growth cycles.

Supplemental reference data will be gathered from local records, higher-resolution aerial maps, and existing mangrove mapping projects. Key sources include Global Mangrove Watch, the 2023 Philippine Mangrove Map, and records from the Aringay Municipal Environment and Natural Resources Office (MENRO). These sources will serve as guidance for building the reference dataset and evaluating the final map outputs, rather than being assumed as error-free baseline truth.

Image Preprocessing. In the second phase, the raw Sentinel-2 imagery will be prepared for model training. The exact same processing steps will be applied to every observation year: removing clouds and cloud shadows, cropping the images to the study area boundary, aligning spatial grids, resizing lower-resolution bands, normalizing values, handling missing pixels, and calculating the NDVI layer.

The model will process five distinct image layers: Sentinel-2 Band 3 (Green), Band 4 (Red), Band 8 (Near-Infrared), Band 11 (Short-Wave Infrared), and the Normalized Difference Vegetation Index (NDVI). Adding NDVI supplies explicit vegetation health information alongside the raw color channels. Rather than assuming this five-layer set is automatically optimal, its performance will be formally tested. Combining raw color layers with vegetation indicators aligns with established methods in mangrove mapping literature (Maung et al., 2024; Xue & Qian, 2022).

NDVI will be calculated using the standard formula:

$$NDVI = (B_8 - B_4) / (B_8 + B_4)$$

While Bands 3, 4, and 8 have a 10m spatial resolution, Band 11 naturally has a lower resolution of 20m. To combine all layers into a unified stack, Band 11 will be resampled to match the 10m grid using bilinear interpolation. This step aligns the spatial grid across all layers without inventing new detail beyond Band 11's original resolution.

The full satellite images will be divided into 256 x 256 pixel patches. All input layers and their corresponding target reference masks will share the exact same grid alignment, ensuring that every pixel directly matches its true ground category.

Dataset Labeling and Partitioning. The third phase will involve preparing reference masks that show mangrove and non-mangrove areas. Mangrove pixels will be given a value of one (1), while non-mangrove pixels will be given a value of zero (0).

One main reference source will be used as the starting point for preparing the masks. Other available datasets, local records, and high-resolution images will be used to check unclear areas. Special attention will be given to small or separated mangrove areas, narrow coastal strips, areas where mangroves meet water, and areas where mangroves change into other types of land. If the reference sources show different results, the study will follow set rules to decide the correct label. Areas that cannot be clearly identified will be recorded and may be left out of model testing.

Reference labels will be prepared through the interpretation of available reference datasets, supporting local records, and appropriate high-resolution imagery. To improve the reliability of the reference labels, the initial labeling produced by the researchers will be reviewed and validated by the core officers of the Aringay Municipal Environment and Natural Resources Office (MENRO), who will serve as the study's designated validators independent from the researchers who prepared the initial labels. This expert review will

focus on confirming the correct classification of ambiguous or difficult areas, including fragmented mangrove stands, narrow coastal strips, and boundaries between mangroves and adjacent land or water cover. Labels flagged as uncertain or inconsistent during this review will be revised prior to finalization.

Before training the model, the dataset will be divided into 80% for training and 20% for validation. The training set will be used to train the U-Net model, while the validation set will be used to check how well the model works on images that were not used during training.

U-Net Model Training. The fourth phase will focus on developing, training, checking, and selecting the U-Net model for mangrove mapping. U-Net, introduced by Ronneberger et al. (2015), is a deep learning model that can identify and label each pixel in an image. Its design allows it to capture both the important features and locations of mangrove areas, making it suitable for creating detailed mangrove maps.

U-Net Architecture. U-Net will be the main deep learning model used in this study. It has several parts that work together to identify mangrove areas in an image. The first part identifies important features from the image, the middle part processes these features, and the last part uses them to create a clear map of the mangrove areas. Connections between these parts help keep important details, especially around mangrove boundaries and small or separated mangrove areas.

U-Net is suitable for this study because the goal is to identify mangrove and non-mangrove areas pixel by pixel, rather than classifying the whole image as one type. Previous studies have used U-Net and similar models for mangrove mapping using Sentinel-2 and other satellite images (Maung et al., 2024; Jamaluddin & Chen, 2024;

Auxtero & Villamor, 2026). These studies support the use of U-Net for this study. However, their results will not be assumed to be the same for the study area. Instead, the model will be tested based on the specific environmental and location conditions of Barangay Dulao.

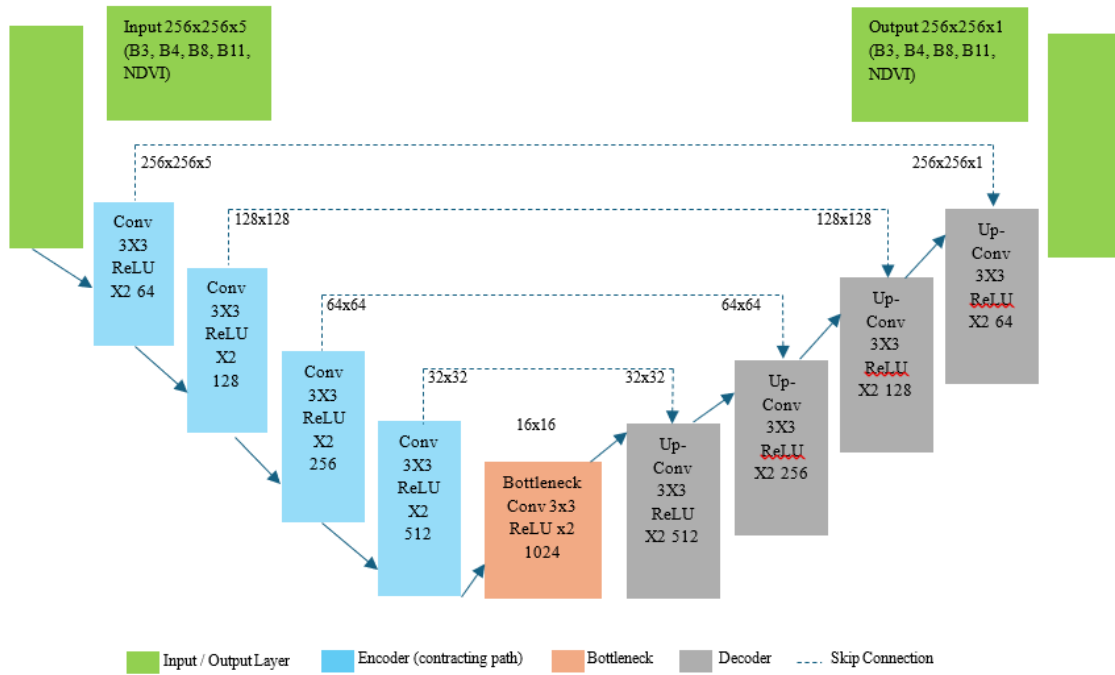


Fig. 1. U-Net Architecture used in this study, adapted from the encoder-decoder design of Ronnenberger et al. (2015) configure for a 5-channel (Bands 3, 4, 8, 11, NDVI) input at 256x256 resolution.

As illustrated in Figure 1, the proposed U-Net will receive five-channel image patches consisting of Sentinel-2 Bands 3, 4, 8, and 11 together with NDVI. The encoder will progressively extract features from the input, while the decoder will reconstruct the spatial representation through upsampling and the integration of encoder features through skip connections. The final output layer will produce a single-channel probability map representing the estimated probability of each pixel belonging to the mangrove class.

The architecture of the model will include encoder, bottleneck, decoder, skip connections, output. The five-channel input patches acquired from Sentinel-2 will be fed

to an encoder. The Bottleneck continues to compress the feature information, which the decoder restores spatially. Skip connections will transport spatial information from corresponding encoder layers to decoder layers to help maintain mangrove boundaries and spatial details.

The input layer will receive image patches with dimensions of 256×256 pixels and five channels. The five channels will consist of Sentinel-2 Bands 3, 4, 8, and 11 and the derived NDVI layer. These inputs will provide complementary spectral and vegetation information for distinguishing mangrove from non-mangrove pixels.

The output layer will use a sigmoid activation function to produce a $256 \times 256 \times 1$ probability map. Each output pixel will have a value ranging from 0 to 1, representing the estimated probability that the corresponding pixel belongs to the mangrove class. The probability map will subsequently be converted into a binary mangrove/non-mangrove segmentation mask using a classification threshold determined from the validation dataset.

The model will be compiled with an Adam optimizer along with a 0.001 learning rate. The pixel-wise binary classification of mangrove and non-mangrove classes makes the binary cross-entropy the primary loss function for this model.

The training process will record the batch size, maximum number of epochs, learning rate, augmentation procedures, class-imbalance management and stopping criteria. While developing the model, the training loss and validation loss will be monitored for overfitting or underfitting. The candidate final model will be the model configuration that resulted in the best validation performance as per the prescribed selection process.

Using the validation dataset, the classification threshold will be selected. Once the threshold and final model configuration have been selected, both will be fixed before the independent test evaluation and multi-temporal application.

Model Validation and Final Testing. The fifth phase uses the validation subset to evaluate the candidate U-Net models. Model performance will be assessed using Intersection over Union (IoU), F1-score, and mean Average Precision (mAP).

F1-score will be calculated as:

$$F1\text{-score} = 2TP / 2TP + FP + FN$$

IoU will be calculated as:

$$IoU = TP / TP + FP + FN$$

where TP represents true-positive pixels, TN true-negative pixels, FP false-positive pixels, and FN false-negative pixels. The mangrove class will be treated as the positive class, and all metrics will be calculated at the pixel level for mangrove and non-mangrove classifications.

In addition to the threshold-based metrics, mean Average Precision (mAP) will be computed from the model's continuous probability map. The classification threshold will be swept from 0.05 to 0.95 in increments of 0.05. At each threshold, predictions and reference labels form a confusion matrix from which a precision–recall pair is derived, producing a precision–recall curve across all thresholds. Average Precision (AP) is computed as the area under this curve:

$$AP = \sum (R_n - R_{n-1}) \times P_n$$

where R_n is the recall and P_n is the precision at the n th threshold, with $R_0 = 0$. Because the segmentation task contains a single positive class (mangrove), mAP is

equivalent to the AP of the mangrove class. The precision–recall curve and the AP value will be reported. The same threshold sweep will be used to select the operating classification threshold (the threshold maximizing validation IoU), which will then be locked together with the final model.

During validation, the researchers will evaluate model performance using these metrics and visually inspect the predicted segmentation masks. Attention will be given to continuous mangrove formations, fragmented patches, narrow coastal strips, and mangrove-water boundaries.

If a candidate model does not meet the predefined performance criteria, the appropriate previous phase will be revisited. Label inconsistencies may require revisions to Phase 3, preprocessing issues may require revisions to Phase 2, and model or training-parameter issues may require another iteration of Phase 4.

Once the final model configuration and classification threshold have been established, the model will be evaluated once using the independent test subset. Test results will be reported separately from validation results and will not be used to further modify the model, training parameters, preprocessing procedure, or classification threshold.

Multi-Temporal Change Detection. After final testing, the finalized U-Net model will be applied to the preprocessed Sentinel-2 images for each observation year from 2019 to 2024. The same model configuration, five-channel input structure, preprocessing procedure, patch size, and classification threshold will be used consistently across all years.

Annual binary mangrove/non-mangrove maps will be generated for 2019, 2020, 2021, 2022, 2023, and 2024. Consecutive-year comparisons will then be conducted for:

- 2019–2020;

- 2020–2021;
- 2021–2022;
- 2022–2023; and
- 2023–2024.

The annual maps will be compared pixel by pixel to identify four transition categories:

Table 1. Mangrove Cover Transition Categories Based on Consecutive-Year Map Comparisons

Earlier Year	Later Year	Transition Category
Non-mangrove	Mangrove	Mapped mangrove gain
Mangrove	Non-mangrove	Mapped mangrove loss
Mangrove	Mangrove	Stable mangrove
Non-mangrove	Non-mangrove	Stable non-mangrove

The mapped gain of mangroves will thus correspond to pixels classified as non-mangrove in the previous year and mangrove in the subsequent year. The mapped loss of mangroves refers to pixels that were classified as mangrove in the previous year and non-mangrove in the following year.

Because the final classified imagery will use a 10-meter spatial grid, one fully covered pixel will represent 100 square meters. Mangrove-cover area, gain, and loss will be calculated using:

Because the imagery uses a 10m grid, one pixel equals 100 square meters. Area will be calculated in hectares using:

$$Area_{ha} = N \times 100 \text{ m}^2 / 10,000 \text{ m}^2 / ha$$

or equivalently:

$$Area_{ha} = N / 100$$

where N represents the number of classified pixels.

The outputs of the change detection will include annual mangrove-cover maps, mapped gain-and-loss maps for consecutive years, and corresponding area statistics in hectares. Finally, a general 2019–2024 comparison will also be generated to summarize the overall mapped changes during the observation period.

Detected transitions will be interpreted as mapped mangrove-cover changes and will not automatically be considered actual ecological changes. Possible causes, such as restoration, replanting, natural regeneration, erosion, harvesting, storm damage, tidal conditions, phenological differences, or classification errors, will only be discussed when supported by additional evidence.

Prototype System Implementation. Phase seven involves building the prototype image processing system. This system will integrate the finalized U-Net model to automatically execute the preprocessing, yearly segmentation, change detection, and area computation steps outlined in the previous phases. The prototype creates maps of annual mangrove cover for the selected years of observation based on processed Sentinel-2 images.

The generated maps, tables, area statistics, and summary information will be presented by the system to the possible users, which in this case is the Aringay MENRO.

The prototype will focus on the core functionality of the output interpretability and not operationalization. It will be deemed functionally complete when it can effectively

process the input images, create mangrove segmentation masks, determine mapped cover changes and display the outputs to the user.

Prototype System Evaluation. The eighth phase will evaluate the completed prototype after integration of the finalized U-Net model and multi-temporal change-detection procedures. This phase will focus on functional system performance rather than serving as another model-training or model-selection stage.

The prototype will be evaluated on its ability to: (1) accept prepared Sentinel-2 imagery, (2) execute preprocessing and segmentation, (3) generate annual maps, (4) identify mapped gains and losses, (5) calculate areas correctly, and (6) display maps and summaries accurately. Area calculations will be verified by manually recalculating selected results from the pixel counts. Where local records are available, changes will be examined for context, but this will not serve as a new model validation test.

The principal iteration will occur among Phases 3, 4, and 5 until the validation metrics stop improving. Once finalized, the workflow is considered complete when: (a) all Sentinel-2 imagery and reference data are acquired; (b) images are preprocessed into 5-channel, 256x256 patches; (c) reference masks are quality-checked; (d) the data is split 80/20; (e) the U-Net model is trained and the best configuration is selected; (f) annual maps for 2019–2024 are generated; (g) consecutive-year gains and losses are mapped and quantified in hectares; and (h) the prototype successfully integrates all processing, mapping, and reporting functions.

Data Gathered

The study will gather the datasets and information needed to develop, train, and validate the proposed system. The primary data will be Sentinel-2 multispectral imagery covering Barangay Dulao, Aringay, La Union, from 2019 to 2024. The input data will include Band 3 (green), Band 4 (red), Band 8 (near-infrared), and Band 11 (short-wave infrared), along with the Normalized Difference Vegetation Index (NDVI) calculated from Bands 8 and 4. These bands provide spectral information for distinguishing vegetation, water, and coastal features, a method supported by previous studies (Farzanmanesh et al., 2024; Maung et al., 2024; Xue & Qian, 2022). Reference labels will be prepared as binary masks representing mangrove and non-mangrove areas. These labels will be created using available mangrove datasets, local information, and high-resolution imagery.

The initial reference labels will be produced by the researchers through interpretation of available mangrove datasets, local records, and high-resolution imagery. However, since the researchers are also responsible for developing and training the U-Net model, they will not serve as the sole authority for confirming the correctness of these labels. To avoid this conflict of interest and ensure the reliability of the reference dataset, the researcher-prepared labels will be independently validated by the core officers of the Aringay Municipal Environment and Natural Resources Office (MENRO), who possess direct local knowledge of the municipality's mangrove areas.

The validation process will involve a structured review of the researcher-prepared reference masks by MENRO's core officers, with particular attention given to areas that are difficult to classify, including fragmented mangrove stands, narrow coastal strips, mangrove-water boundaries, and zones transitioning into other land-cover types. Labels

confirmed by the validators will be retained as the final reference masks used for model training and evaluation. Labels flagged by the validators as incorrect, unclear, or inconsistent with on-the-ground conditions will be revised by the researchers and, where necessary, re-submitted for confirmation before being finalized. This validation step will be conducted prior to model training to ensure that the dataset used to train and evaluate the U-Net model reflects locally verified ground conditions rather than the researchers' interpretation alone.

Areas with narrow, fragmented, or recently changed mangrove cover will be cross-checked with higher-resolution imagery to ensure the accuracy of the final reference masks. Additional geographic information about Barangay Dulao will be gathered to define the study area and aid in interpreting the resulting maps. To minimize classification errors caused by seasonal changes or cloud cover, images will be selected from similar times of the year and screened for cloud contamination (Valderrama-Landeros et al., 2021).

Data Analysis

The analysis will begin by training a U-Net model to perform binary semantic segmentation, classifying each pixel as either mangrove or non-mangrove. The labeled dataset will be split into an 80% training set and a 20% validation set. The training set will update the model's weights, while the 20% validation set will serve as the primary estimate of model performance and guide the selection of hyperparameters and the classification threshold. Model performance will be evaluated using F1-score, Intersection over Union (IoU), and mean Average Precision (mAP). These metrics assess the pixel-level agreement

between the predicted masks and the reference masks (Maung et al., 2024; Auxtero & Villamor, 2026).

F1-score will be calculated as:

$$F1\text{-score} = 2TP / 2TP + FP + FN$$

IoU will be calculated as:

$$IoU = TP / TP + FP + FN$$

where TP represents true-positive pixels, TN represents true-negative pixels, FP represents false-positive pixels, and FN represents false-negative pixels.

In addition to the threshold-based metrics, mean Average Precision (mAP) will be computed from the model's continuous probability map. The classification threshold will be swept from 0.05 to 0.95 in increments of 0.05. At each threshold, predictions and reference labels form a confusion matrix from which a precision–recall pair is derived, producing a precision–recall curve across all thresholds. Average Precision (AP) is computed as the area under this curve:

$$AP = \sum (R_n - R_{n-1}) \times P_n$$

where R_n is the recall and P_n is the precision at the n th threshold, with $R_0 = 0$. Because the segmentation task contains a single positive class (mangrove), mAP is equivalent to the AP of the mangrove class. The precision–recall curve and the AP value will be reported. The same threshold sweep will be used to select the operating classification threshold (the threshold maximizing validation IoU), which will then be locked together with the final model.

The performance of the very first trained model will serve as a baseline. Subsequent model iterations will be compared against this baseline to determine if adjustments such as data augmentation, preprocessing tweaks, or threshold changes actually improved the model. This baseline-driven approach is a study-specific evaluation method rather than a universal performance threshold.

Once the final model and classification threshold are established using the training and validation data, they will be locked. This finalized model will then be applied consistently to all preprocessed Sentinel-2 imagery from 2019 to 2024 to generate annual mangrove maps.

For the change detection analysis, the annual maps will be compared pixel-by-pixel for consecutive years (2019–2020, 2020–2021, 2021–2022, 2022–2023, and 2023–2024) to identify mapped mangrove gain, mapped mangrove loss, and stable areas. Because the final classified imagery uses a 10-meter spatial grid, each pixel represents 100 square meters. The area of each transition category will be calculated in hectares using:

$$Area_{ha} = N \times 100 \text{ m}^2 / 10,000 \text{ m}^2 / ha$$

or equivalently:

$$Area_{ha} = N / 100$$

where N represents the number of classified pixels within the corresponding mangrove-cover category.

Finally, the analysis will involve a visual inspection of the mapped change areas. It is important to note that a mapped change in land-cover class does not automatically establish a real-world cause. Changes can result from restoration efforts, natural regeneration, erosion, development, storm damage, tidal variation, phenological changes,

or classification error. Any discussion regarding the actual causes of mangrove gain or loss will only be made when supported by external local records or evidence.

Ethical Consideration

The researchers will observe ethical principles throughout the study, particularly in the collection, processing, analysis, and reporting of geospatial and reference data. The study will primarily use Sentinel-2 multispectral imagery and existing reference datasets for developing and evaluating the U-Net-based mangrove segmentation model. Proper acknowledgment and citation will be provided for all external datasets, satellite imagery, records, and published materials used. Any local records or geographic information obtained from relevant offices will only be used for purposes related to the study and will be handled according to applicable access, use, and confidentiality requirements. The core officers of the Aringay MENRO who will serve as dataset validators will participate voluntarily, with their role, time commitment, and contribution clearly explained and acknowledged in the study.

The researchers will maintain research integrity by following the established procedures for reference-mask preparation and model evaluation. Reference labels will not be modified solely to improve model performance, and ambiguous areas will be appropriately documented or excluded from evaluation when necessary. The researchers will also report the model's results and limitations honestly, including possible effects of reference-data quality, spatial resolution, image conditions, and classification errors.

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APPENDICES

APPENDIX A

Letter to the Adviser




DON MARIANO MARCOS MEMORIAL STATE UNIVERSITY
 South La Union Campus, Agoo, La Union Philippines
COLLEGE OF COMPUTER SCIENCE
 www.dmmmsu.edu.ph | +63 917-728-0563 | ccs.sluc@dmmmsu.edu.ph

July 3, 2026

FERNAN H. MENDOZA
 Faculty Member, College of Computer Science
 DMMMSU-SLUC
 Agoo, La Union

Sir/Madam:

We, the undersigned fourth-year students of Bachelor of Science in Computer Science, are currently enrolled in Thesis Writing 1. As a requirement in the course, we would like to conduct a study about:

- KALINGA: A Gender-Responsive Student Welfare Portal with CNN-Based Document Forgery Detection and Taglish-Aware NLP Harassment Classification
- GABAY-AI: A Retrieval-Augmented Generation Conversational Agent on Philippine Gender Laws for Women's Rights and Crisis Referral
- AQUAVEIL: An IoT-Based Decision Support System for Early Dissolved Oxygen and pH Crash Warning in Freshwater Fishpond Aquaculture

In this regard, may we request you to be our Thesis Adviser. We believe your expertise in the field of Computer Science will provide us with valuable guidance and insights to ensure the success and quality of our study.

We would deeply appreciate if you would consider our request.

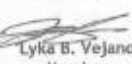
Thank you very much.

Respectfully yours,


Nash Francis M. Caluza
 Leader


Reymark O. Boado
 Member


Eliza Praise Y. Milana
 Member


Lyka B. Vejano
 Member

Approved:


FERNAN H. MENDOZA
 Thesis Adviser









APPENDIX B

Letter of Intent

July 22, 2026

HON. BENJAMIN O. SIBUMA
Municipal Mayor
Municipality of Aringay, La Union



Attn.: **ENGR. RHEA MAE B. TUGADE**
OIC - Municipal Environment and Natural Resources Officer (MENRO)

Subject: Letter of Intent for Thesis Collaboration and Request for Institutional Support

Dear Sir:

Greetings!

We are fourth-year Bachelor of Science in Computer Science students from Don Mariano Marcos Memorial State University (DMMMSU) – South La Union Campus. We are currently conducting our undergraduate thesis entitled: **"Deep Learning-Based Detection of Mangrove Gain and Loss in a Local Coastal Site."**

The objective of our study is to develop an image processing system that uses free Sentinel-2 satellite imagery and artificial intelligence (AI) to detect and monitor changes in mangrove forest cover over time. The system aims to identify areas where mangroves have increased or decreased and generate maps and statistics that can support environmental monitoring and decision-making.

In this regard, we respectfully seek the support of your office, particularly through the Municipal Environment and Natural Resources Office (MENRO), for the conduct of our research. Specifically, we kindly request:

- Permission to conduct our study within the municipality's mangrove areas;
- Access to available records related to mangrove planting, rehabilitation, coastal resource management, or similar environmental activities, if available;
- Coordination with the MENRO Office for field validation, if necessary; and
- The issuance of a Commitment Letter signifying your office's willingness to collaborate with us for this academic research.

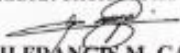
We assure you that all information and records obtained will be used solely for academic and research purposes and will be treated with confidentiality and in accordance with applicable ethical standards.

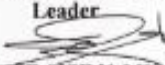
Upon completion of the study, we will provide your office with a copy of our research findings, including a mangrove cover change map, estimated mangrove gain and loss statistics, and other outputs that may assist the municipality in future environmental monitoring and conservation initiatives.

We sincerely hope for your favorable consideration of this request. Should you require additional information, we would be pleased to meet with you or your designated representative at your convenience.

Thank you very much for your time and support. We look forward to the opportunity to collaborate with your office.

Respectfully yours,
The Student Researchers


NASH FRANCIS M. CALUZA
Leader


ELAIZA PRAISE Y. MIANA
Member


REYMARK O. BOADO
Member

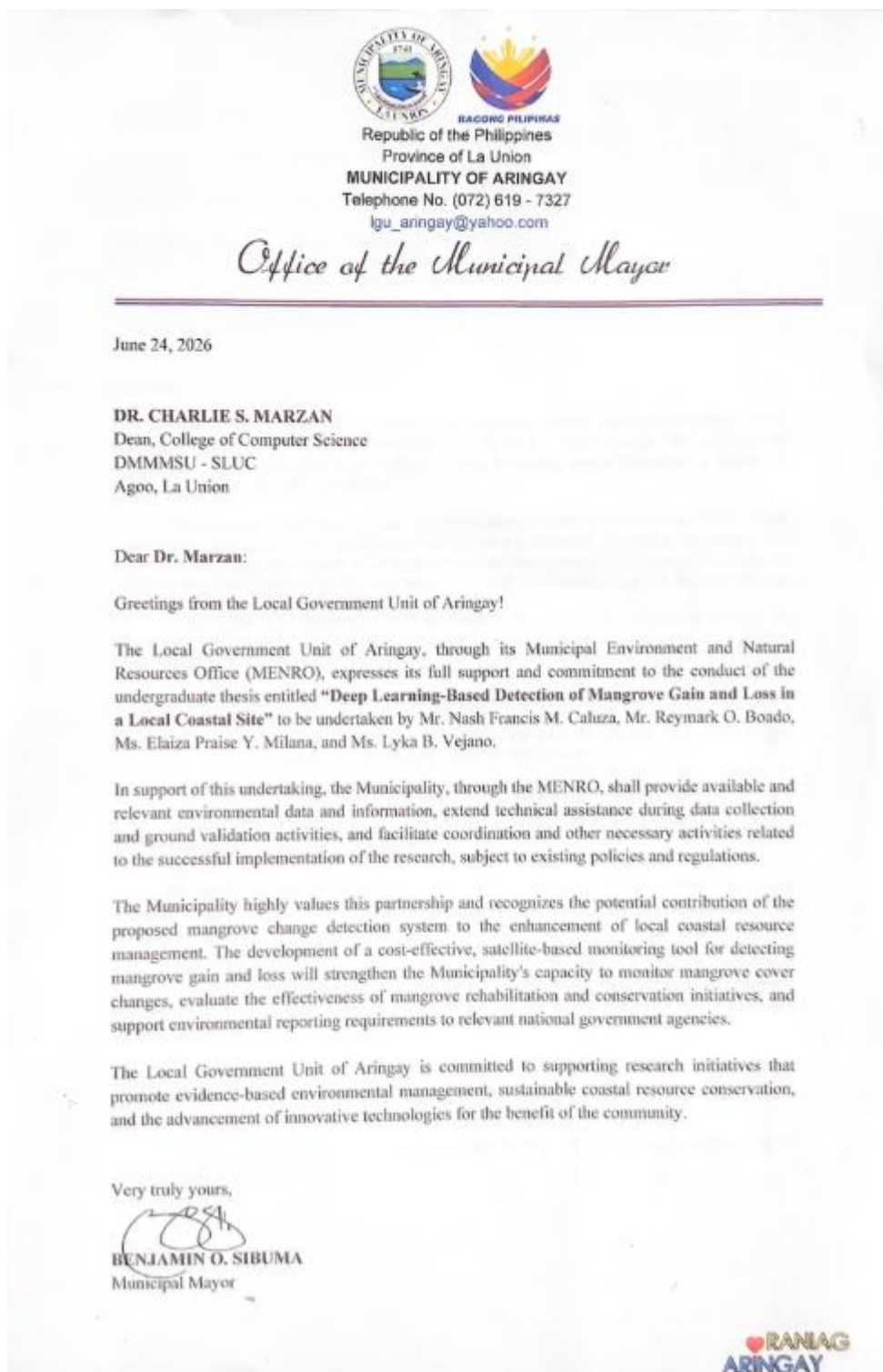

LYKA B. VEJANO
Member

Noted by:


DR. FERNAN H. MENDOZA
Thesis Adviser

APPENDIX C

Letter of Commitment



APPENDIX D
Letter to the Panel Member

CURRICULUM VITAE

NASH FRANCIS M. CALUZA

Poblacion, Aringay, La Union
 2503 Philippines
 Contact No: 09777540959
 E-mail: nashiicaluza@gmail.com



Educational Background

<i>2023 – present</i>	Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union
<i>2021– 2023</i>	Aringay Senior High School Technical Vocational Livelihood- Information Communication Technology
<i>2017– 2021</i>	Notre Dame Institute Poblacion, Aringay, La Union
<i>2011– 2017</i>	Aringay Central Elementary School Poblacion, Aringay, La Union

Seminars and Trainings Attended

<i>2023</i>	Computer Systems Servicing NC II Technical Education and Skills Development Authority
<i>2024</i>	DETECT, DEFEND, AND PROTECT: Introduction to Cybersecurity University of the Philippines Diliman Technology Management Center, Office of the Chancellor
<i>2025</i>	Introduction to Artificial Intelligence (AI) and Machine Learning (ML) for SUCs University of the Philippines System Information Technology Development Center (ITDC)

Special Skills and Talents

- Programming: Java, Python, C / C++ (basic) | Website Development: HTML, CSS, JavaScript, React Native (basic) Tailwind (basic), Figma
- Networking: Local Network Setup, Basic Trouble Shooting, Router Configuration
- General: Document Writing, Communication and Public Relations, Time Management and Adaptability, Leadership

Certified true and correct.

NASH FRANCIS M. CALUZA
Signature over printed name

ELAIZA PRAISE Y. MLANA

San Agustin Sur, Agoo, La Union

2504 Philippines

Contact No: 09464912676

E-mail: elaizapraisemilana@gmail.com

Educational Background

<i>2023 – present</i>	Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union
<i>2021– 2023</i>	Don Eufemio F. Eriguel Memorial National High School Academic- Science, Technology, Engineering, and Mathematics
<i>2017– 2021</i>	Don Eufemio F. Eriguel Memorial National High School Consolacion, Agoo, La Union
<i>2011– 2017</i>	Agoo East Central School Consolacion, Agoo, La Union

Achievements

<i>2024</i>	Philippine Association of State Universities and Colleges (PASUC) REGION I – HIPHOP CHAMPION DMMMSU-MLUC San Fernando City, La Union
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Special Skills and Talents

- Programming: Java, Python, C | Website Development: HTML, CSS, JavaScript, React Native (basic)
- General: Document Writing, Public Relations & Communication, Leadership & Team Management, Adaptability & Time Optimization
- Creative & Technical Skills: Graphic & Layout Design (Photographs, Visuals, Documents), Visual Arts & Drawing
- Performing Arts: Dance

Certified true and correct.

ELAIZA PRAISE Y. MILANA
Signature over printed name

LYKA B. VEJANO

2504 Philippines

San Marcos Agoo, La Union

Contact No: 09509631066

Email: vejanolyka@gmail.com

Educational Background

<i>2023 – present</i>	Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union
<i>2021– 2023</i>	Don Eufemio F. Eriguel Memorial National High School Academic- Humanities and Social Sciences
<i>2017– 2021</i>	Tibagan High School Brgy. East Rembo, Makati City
<i>2011– 2017</i>	Agoo West Central School San Nicolas Norte, Agoo, La Union

Special Skills and Talents

- Basic graphic design and photo editing using Adobe Photoshop
- Document preparation and presentation using Microsoft Office Applications
- Basic UI/UX design and prototyping using Figma

Certified true and correct.

LYKA B. VEJANO
Signature over printed name

REYMARK O. BOADO

Santa Rita Sur, Agoo La Union

2504 Philippines

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Educational Background

<i>2023 – present</i>	Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union
<i>2021– 2023</i>	Don Eufemio F. Eriguel Memorial National High School Academic- Humanities and Social Sciences
<i>2017– 2021</i>	Don Eufemio F. Eriguel Memorial National High School Consolacion, Agoo, La Union
<i>2011– 2017</i>	San Vicente - San Agustin Elementary School San Vicente Norte, Agoo La Union

Special Skills and Talents

- Website Development: React Native (basic)
- Programming: Java and C (basic)
- General: Time management, adaptability

Certified true and correct.

REYMARK O. BOADO
Signature over printed name